Value Streams, Flow, and Product Management

Alex Barsi Lopes
Kelley School of Business

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Agenda

- Value stream management
- Flow
- Managing flow
- Product management

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Value stream is a foundational concept for Lean Thinking

- Lean thinking is based on five related principles
  - Precisely specify value by specific product
  - Identify the value stream for each product
  - Make value flow without interruptions
  - Let the customer pull value from the producer
  - Pursue perfection
- Two main value streams
  - Development value stream
  - Operational value stream

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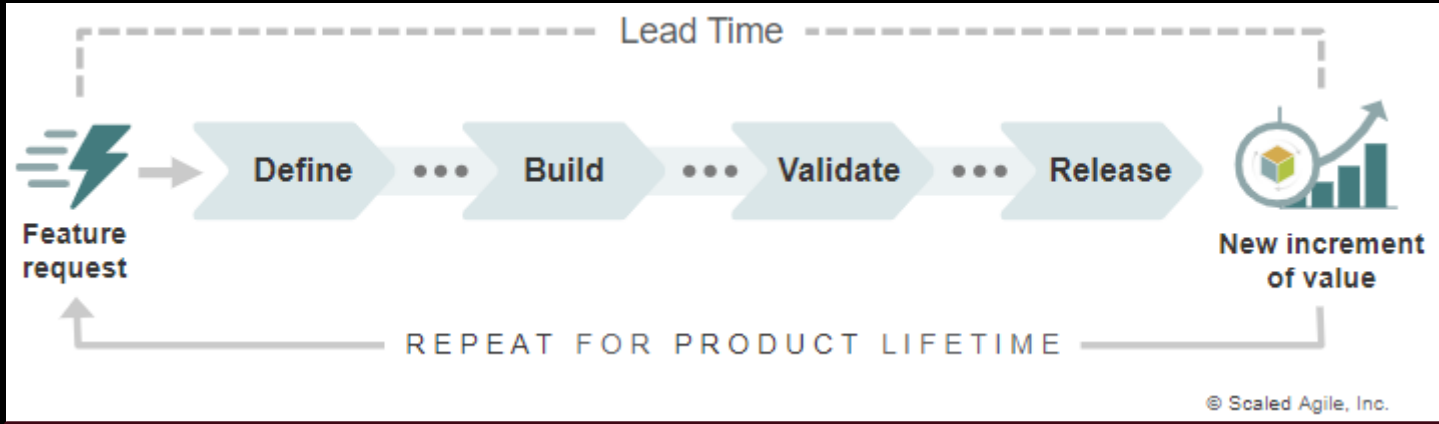
Development value streams are activities related to the development of solutions

- A trigger initiates a new request
- Steps are the activities needed to define, build, validate, and release new value
- The bar between steps indicates the flow of information between steps
- Ellipses indicates delays between steps



- The model above is very, very simplified in comparison to what takes place in modern software development
- The output of the Development Value Stream is a new increment of the solution, the additional value the new features will provide

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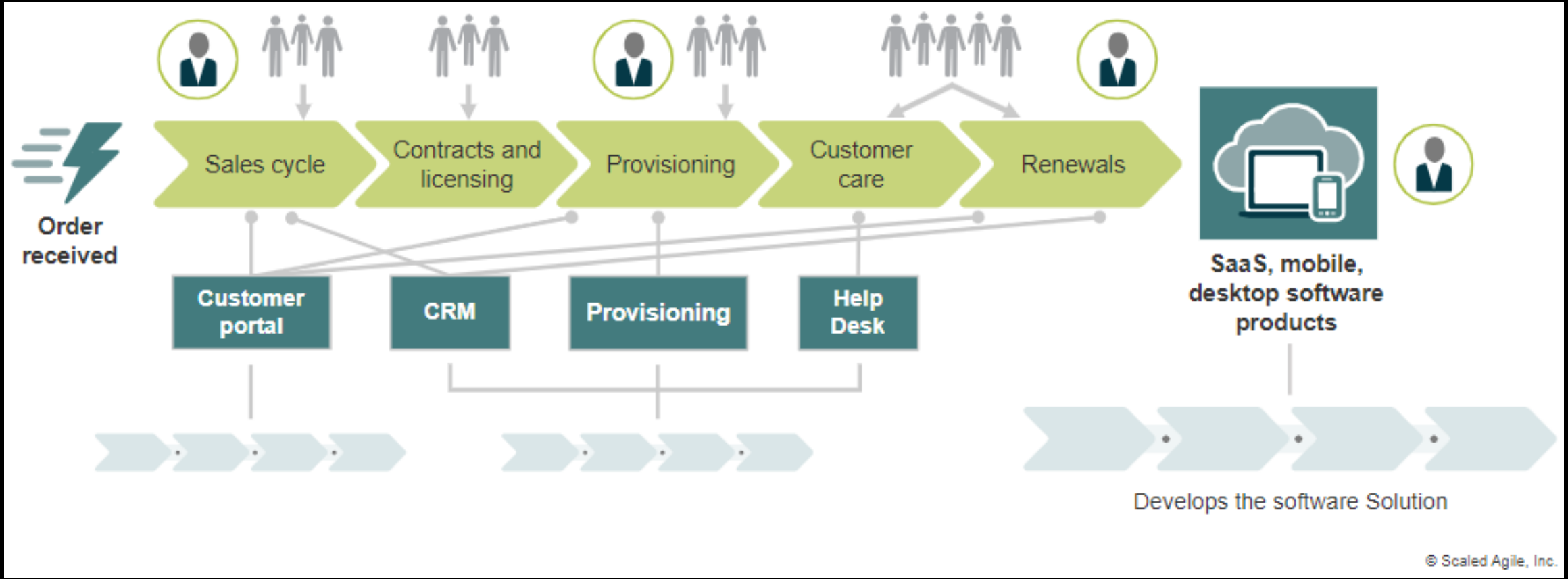
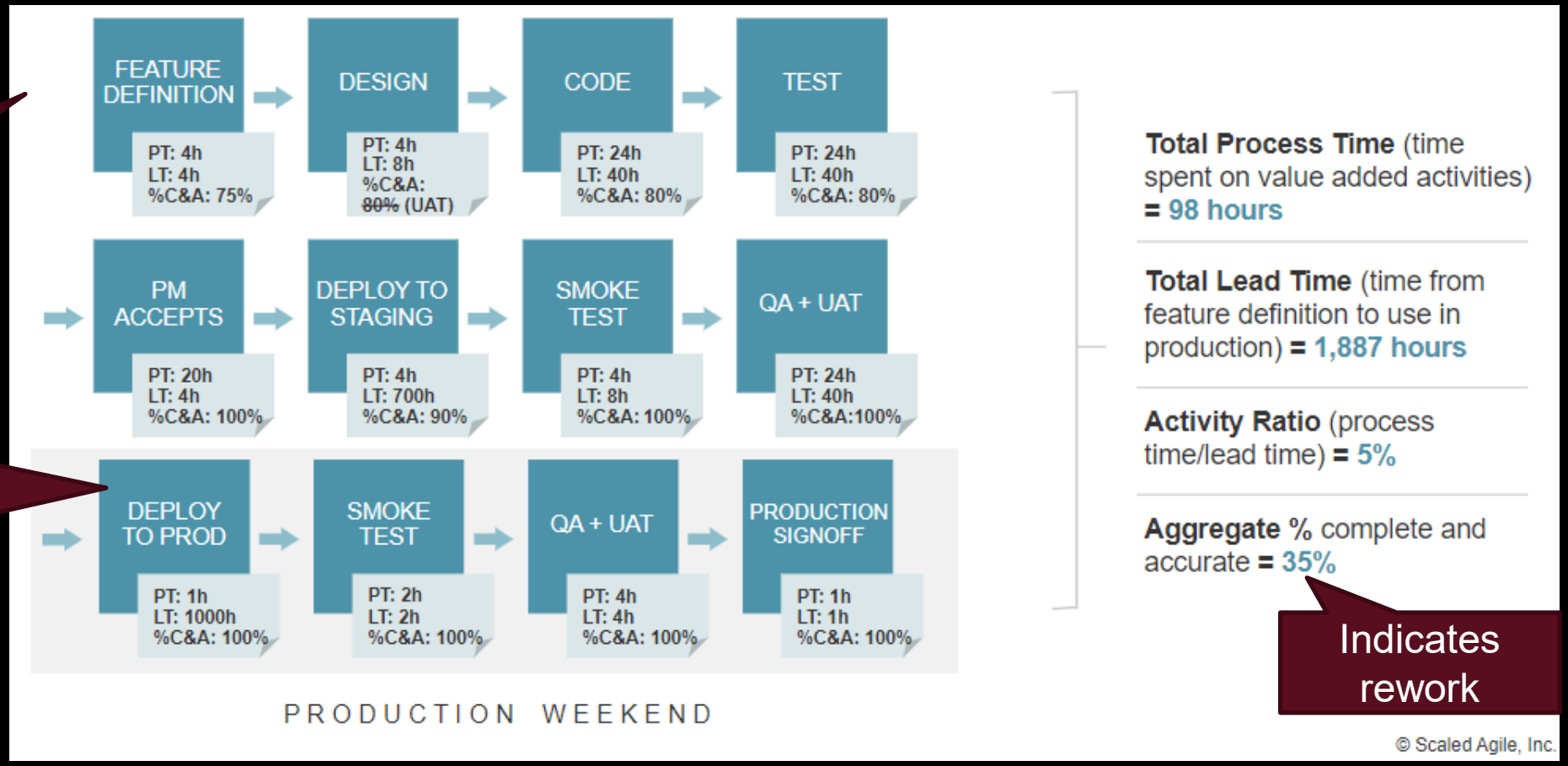
A pattern for the Development Value Stream for software products



Measuring and optimizing development value streams

Sample DVS

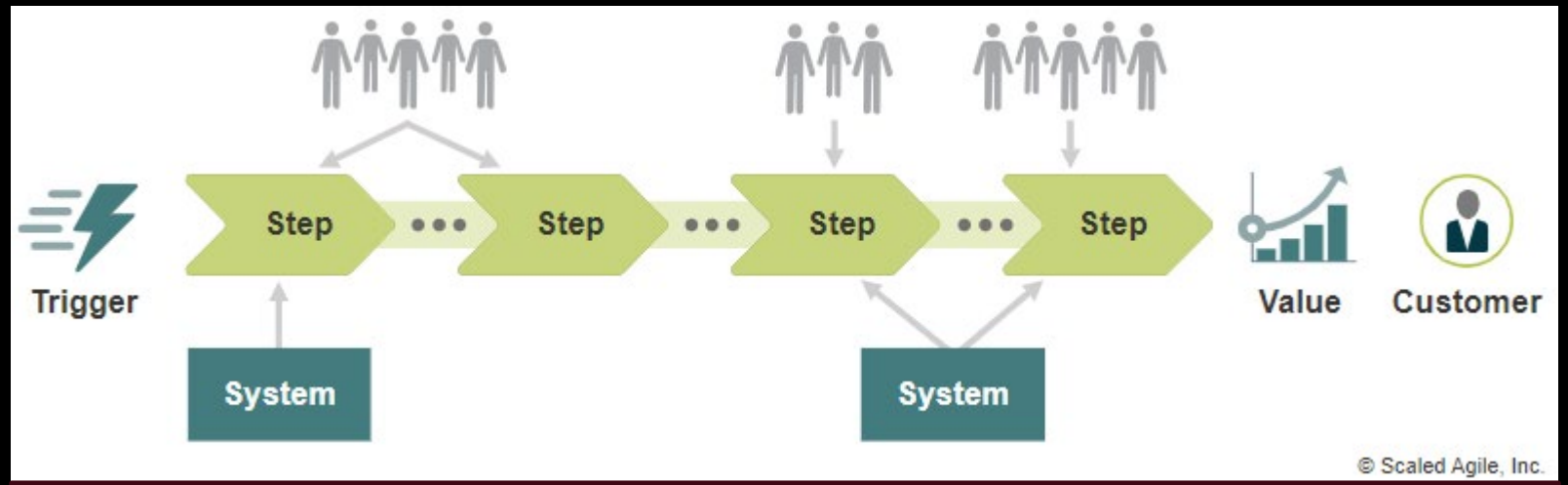
Significant opportunities for improvement, especially with DevOps



Indicates rework

Operational value streams focus on the delivery of a product or service

- Steps necessary to cover the trigger into value delivered
- People performing the steps
- Systems used in the work
- Flow of information and materials to satisfy the request



- Four common Patterns for operational value streams
  - Fulfillment
  - Manufacturing
  - Software products
  - Supporting

Some questions can be used to identify operational value streams

- Who are your users and customers? Are they internal, external, or both?
- What products and services do you market, sell and support?
- What solutions do you provide for internal users?
- What triggers the flow of value?
- What value do your customers perceive you deliver?

With operational value streams, delays are one of the issues that decrease value the most – Minimizing lead time is fundamental

Operational value streams can be matched with customer journey maps

Loan need → Attract customer → Quick rate quote → Complete loan application → Eligibility decision and underwriting → Extend loan terms → Award loan → Setup payment terms → Repay money → Close loan → Loan repayment with interest

Young Couple	DECIDE *Access options*	LEARN *Clarify goals*		CHOOSE *Choose what to buy*		APPLY *ID & financial check*		PURCHASE *Negotiate sale*	COMPLETE *Celebrate*
	Decide to buy a home	Learn about the mortgage	Calculate the budget	Property search	Compare mortgages	Apply for mortgage loan	Assessment	Sign contract & deal	Move
Thinking & Feeling	We loved our neighborhood growing up and want to raise our kids the same way	A real estate agency will save money		Our real estate agent helped us clarify our goals		Just something that will pass quickly	The bank officer will handle everything	We're happy the bank officer kept us updated	
	We've never made a purchase this large before – it is scary!	Visiting bank for consultation = helpful		Selecting real estate agency – comparing & selecting property		Why does the bank need paper copies?		Negotiating a sale this big makes us nervous	We need to clean
		Can we really find a home that we love that we can afford?		Form filing is time consuming & the bank should know our details already		Can't this all be online?		It took a lot longer to close the sale than expected	
Bank		Customer Service Officer • Explains products • Prepares offer • Advises to select real estate agency		Customer Service Officer • Fills in application form & copies ID cards • Provides list with required documents		Credit Department • Performs assessment of applicants • Evaluates property • Prepares approval		Customer Service Officer • Provides updates on loan status	
Technology Opportunities	Online resources for understanding mortgage options	Online advisory, budgeting, financial management, and purchasing options		Automatically gather data from public databases and integrate with social media data		AI-based assessments could improve speed and accuracy		Biometric authentication of electronic signatures	Automated referrals to partners

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- This is an example focusing on a loan process
- There are different templates for customer journey maps
- The focus here is the match with the OVS

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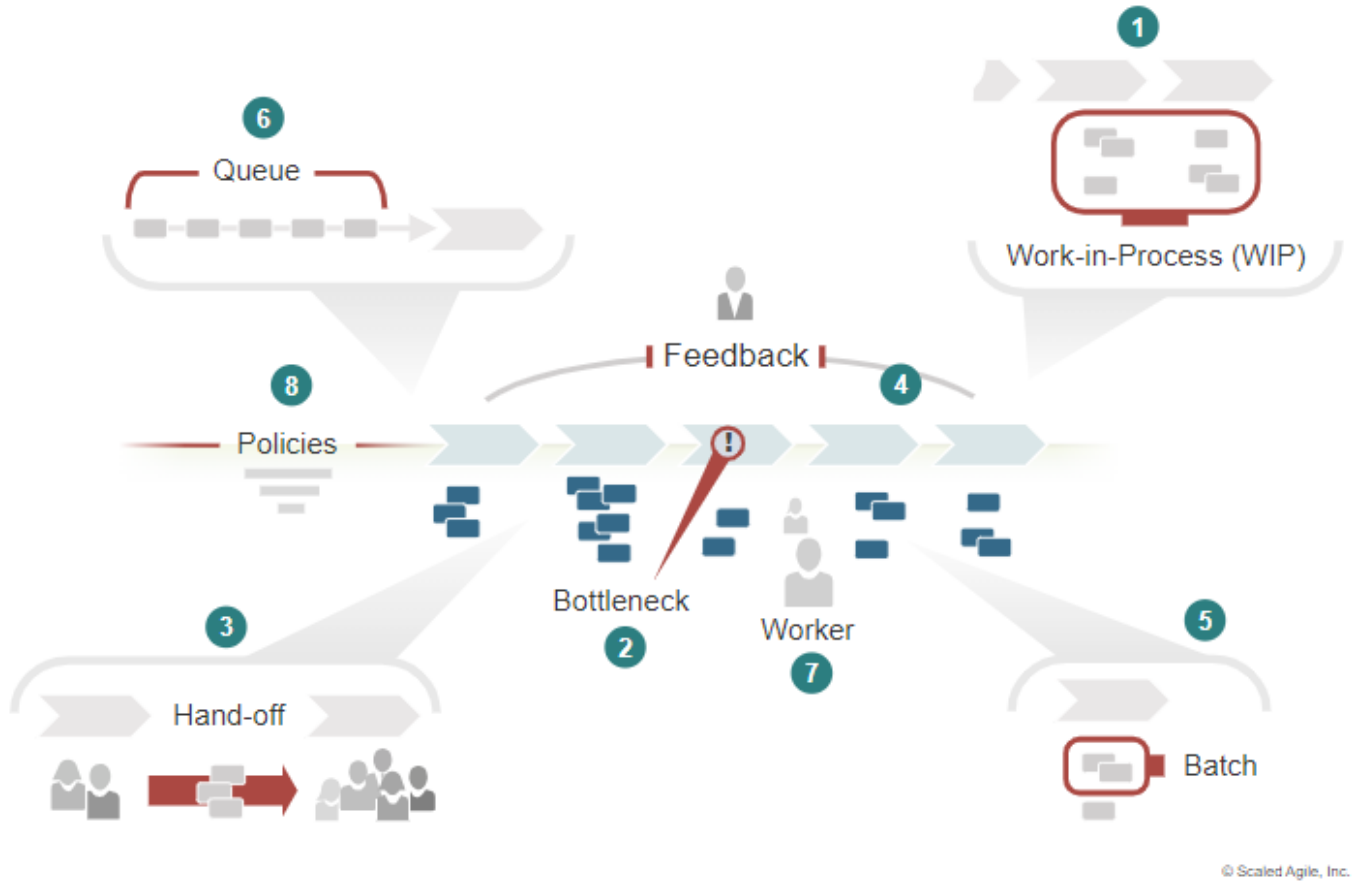
What is flow?

- “Flow occurs when there is a smooth, linear, and fast movement of work product from step to step in a relevant value stream”
- All flow systems have eight common properties



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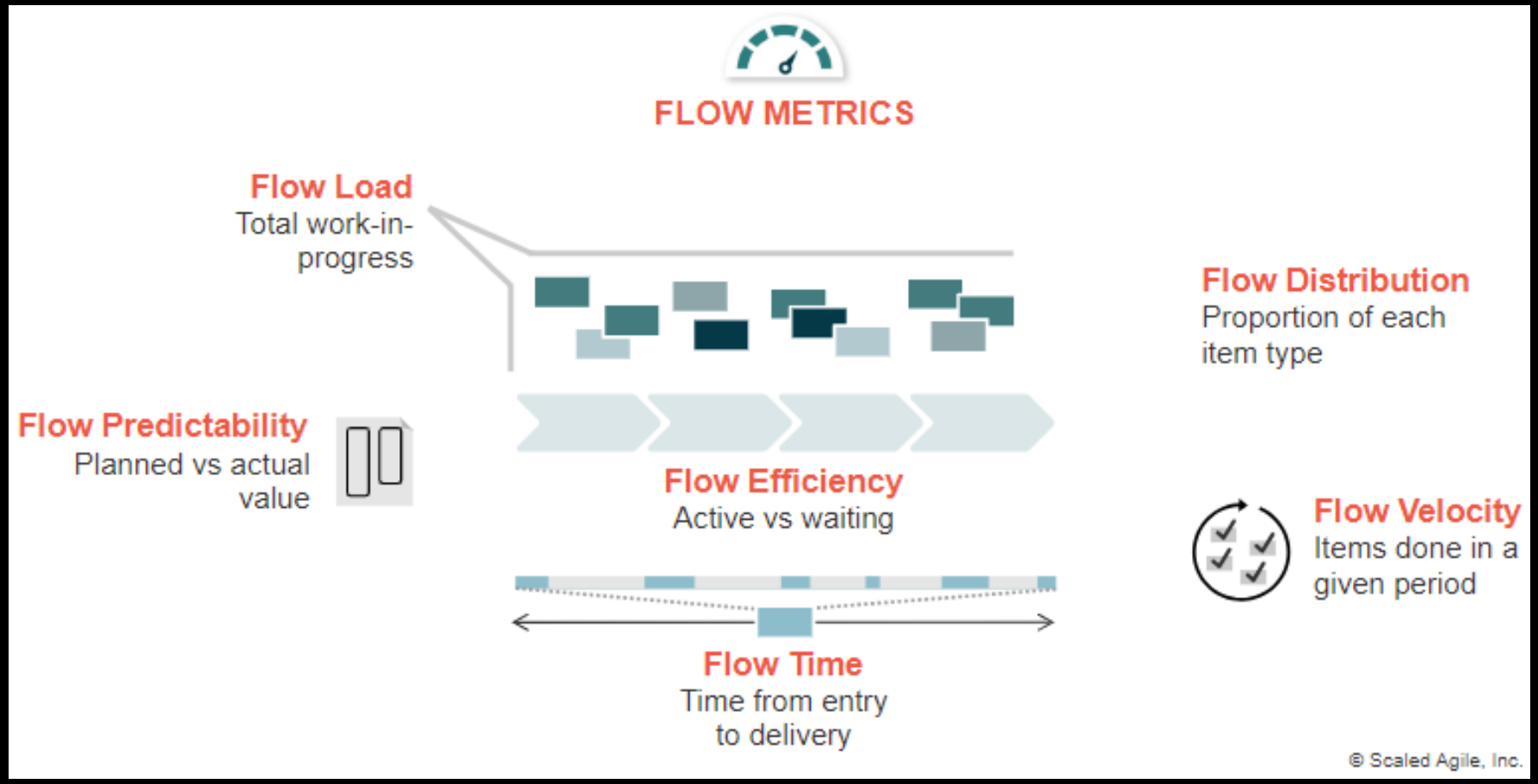
Metrics are used to pursue perfection by improving flow



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There are eight flow accelerators

- Visualize and limit Work-in-Progress (WIP)
- Address Bottlenecks
- Minimize handoffs and dependencies
- Get faster feedback
- Work in smaller batches
- Reduce queue length
- Optimize time “in the zone” (that is, time when people are engaged and productive, working without interruptions)
- Remediate legacy policies and practices

Visualize and limit Work-in-Progress (WIP)

- Too much WIP decreases productivity and limits value delivery
- Actions to be implemented
  - Make current features and WIP visible
  - Balance WIP against managed capacity
  - Ignore sunk costs

Address Bottlenecks

- Bottlenecks will affect team productivity, especially when moving from one step in the process to the next one
- Actions to be implemented
  - Identify the bottleneck and understand its full impact
  - Increase capacity at the bottleneck
  - Bypass the bottleneck when allowed, moving ahead with other features

Minimize handoffs and dependencies

- Excessive and unnecessary dependencies and handoffs disrupt flow and create delays
- Actions to be implemented
  - Make handoffs and dependencies visible and take corrective action when possible
  - Foster incremental execution of dependencies, minimizing their effect
  - Synchronize frequency
  - Optimize the structure of teams and even value streams

Get faster feedback

- When feedback is delayed, we have a higher chance of misunderstanding, rework, slow delivery and unhappy customers
- Actions to be implemented
  - Determine which types of feedback are missing or inadequate
  - Shift reviews left
  - Integrate and test frequently
  - Build quality in
  - Demonstrate working systems
  - Frequently integrate and engage with the customer

Work in smaller batches

- Large batches of work might delay feedback, increase variability, and lead to more rework
- Actions to be implemented
  - Implemented short cadence and relatively small team sizes
  - Adjust the process to support smaller batches
  - Automate a delivery pipeline

Reduce queue length

- The longer the queue, the longer the wait time for the new functionality
- Actions to be implemented
  - Use concise iteration objectives and move additional requests to the backlog
  - Keep roadmaps flexible
  - Establish a strong Product Management function and eliminate non-strategic and bad ideas quickly
  - Leave capacity for emergent priorities

Optimize time “in the zone”

- By minimizing interruptions, contributors can reach a productive state where work is performed better
- Actions to be implemented
  - Optimize meetings and events, and eliminate redundant practices
  - Use innovative collaboration patterns like peer work and mob programming
  - Frequently integrate work to minimize interruptions due to dependencies

Remediate legacy policies and practices

- Legacy policies and practices might impede progress if they do not fit in the new way of doing things
- Actions to be implemented
  - Keep an eye on process and tasks that are redundant or done because “this is how we have done it in the past”
  - Avoid adding traditional waterfall and Project Management techniques that do not fit with a more agile approach

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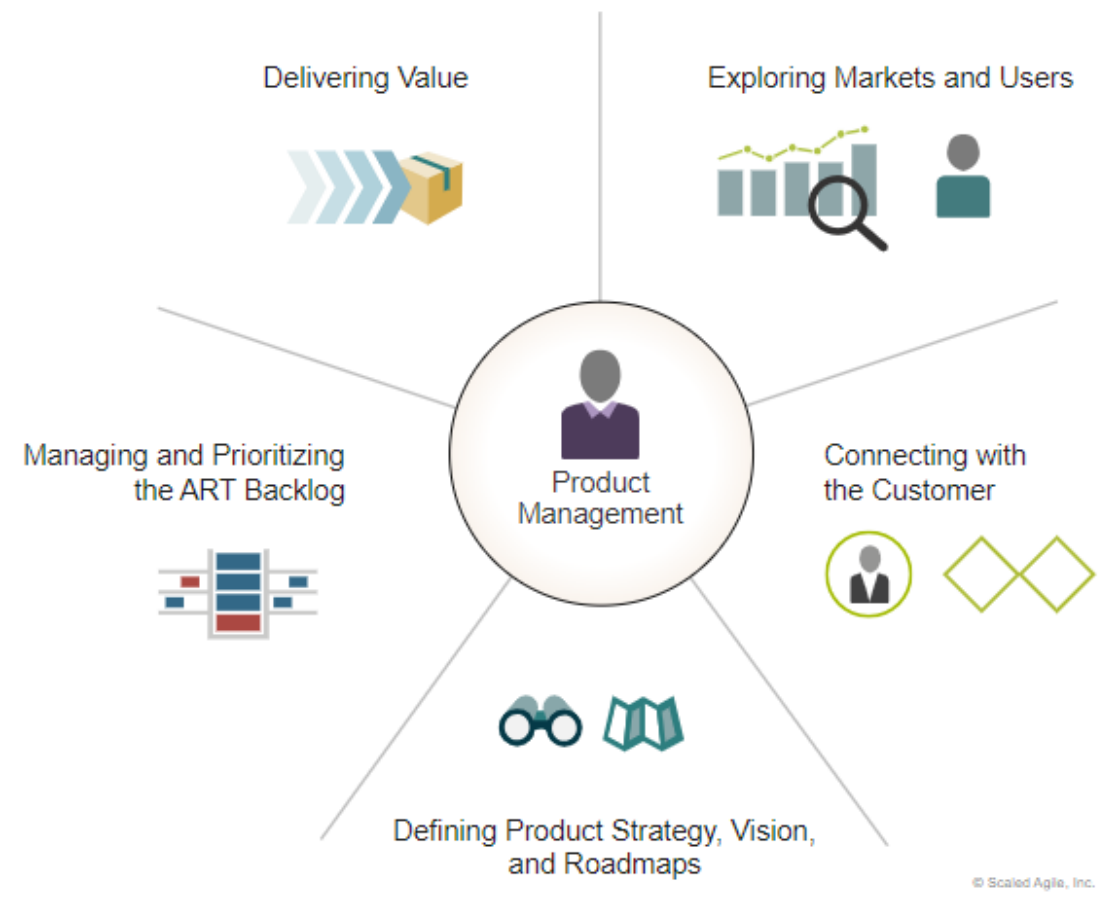
What is Product Management?

- "Product Management is the function responsible for defining desirable, viable, feasible, and sustainable solutions that meet customer needs and supporting development across the product life cycle"
  - Solution management is a scaled-up approach to managing related large-scale products
- Many different managerial techniques are used by Product Managers



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